

Digitalising Social Protection: Better Targeting and Delivery

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► BIO

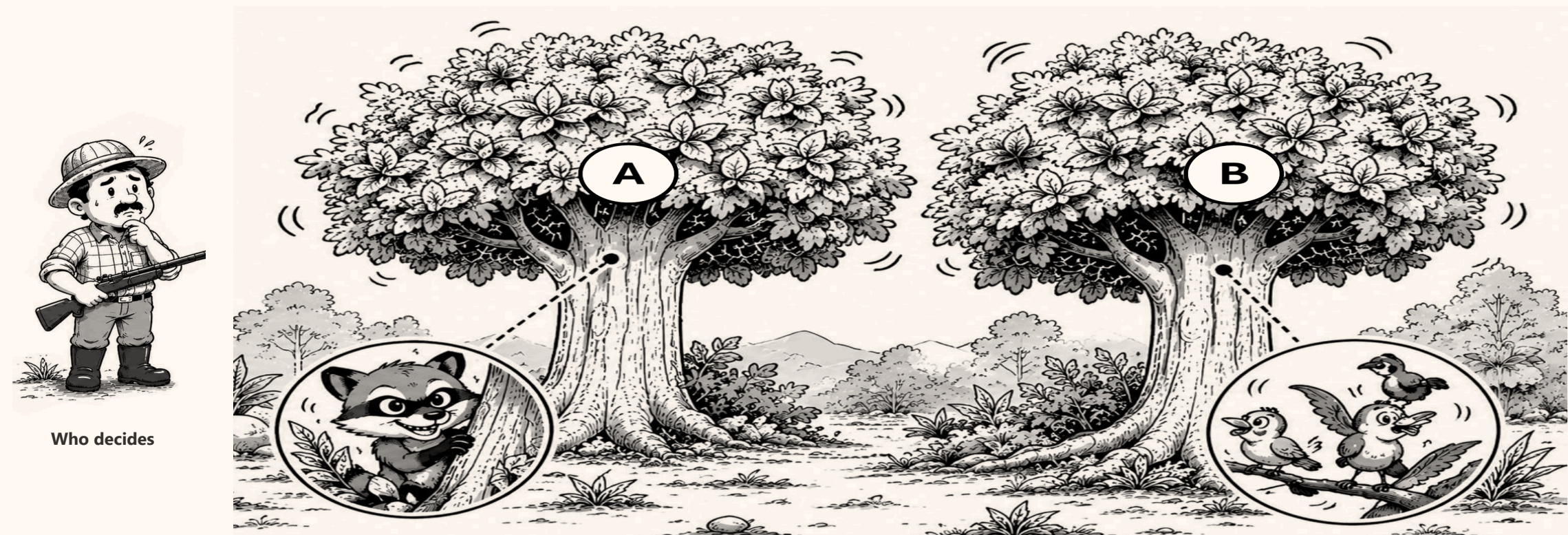
Social assistance in Indonesia misses a large share of the households it is meant for. This asks three questions: how large the gap is, where it comes from, and which parts of it digitalisation can actually close.



THE PROBLEM

We cannot see who is poor. We only see signs.

A hunter cannot see the animal in the tree. He can only see the leaves move. Targeting works the same way: welfare itself is never observed, only proxies for it.



Who decides

Tree A hides the **musang** — the household that qualifies. Tree B holds **birds** — households that look the same from outside.

Exclusion error — the musang escapes: a household that qualifies gets nothing **Inclusion error** — a bird is hit: a household that does not qualify receives the benefit

I How well does it reach people?

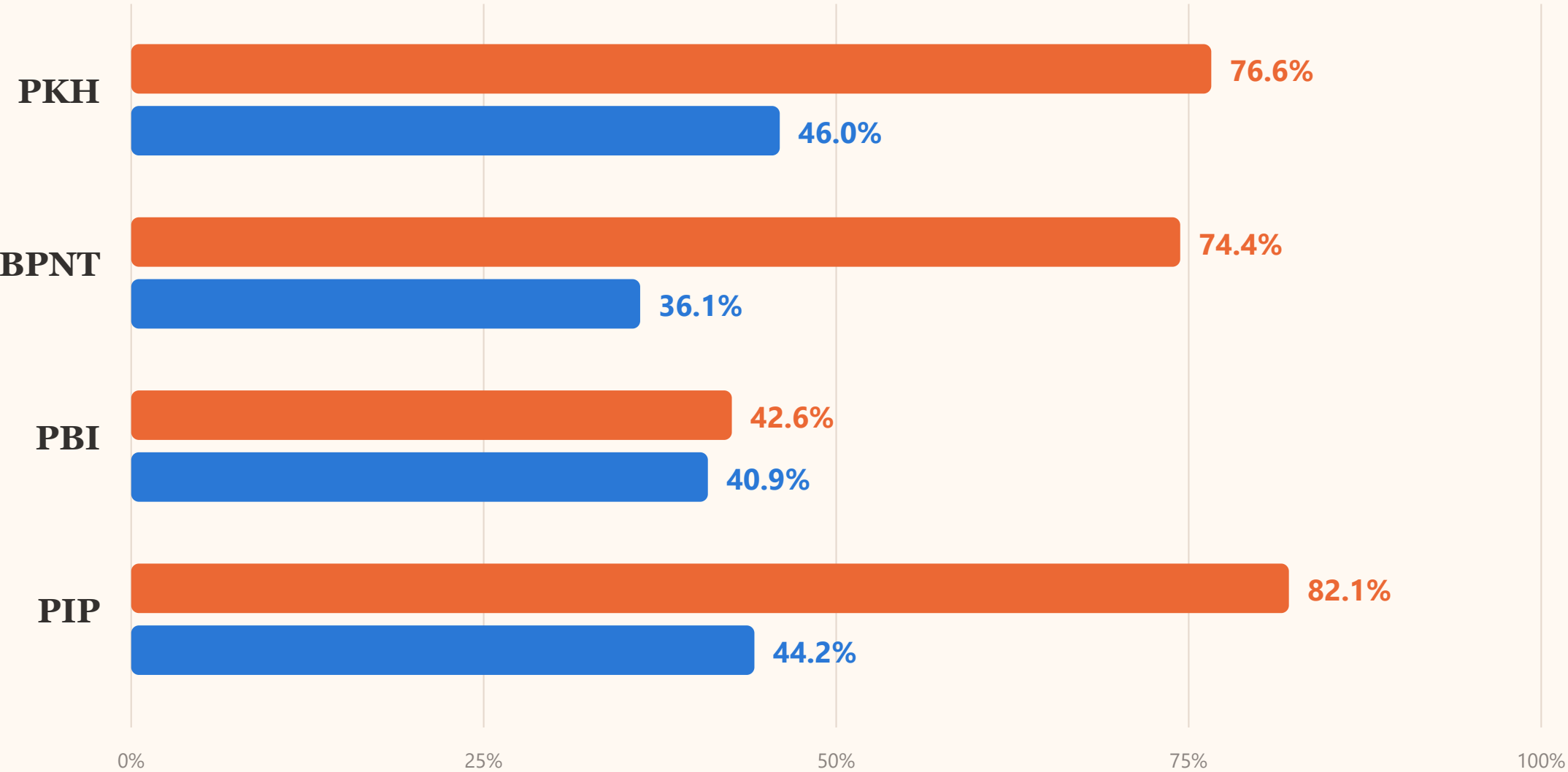
Four programmes, measured against the households and people they are meant to serve.

FIGURE 1

How much of the target group is missed

Susenas March 2025. Exclusion error is the share of the target group that receives nothing. Inclusion error is the share of recipients who fall outside the target group.

Exclusion error — the target group that receives nothing Inclusion error — recipients outside the target group



Move the PKH cut-off to deciles 1–3 and inclusion error becomes 57.7%; move it to 1–5 and it becomes 35.7%.

Nothing about the programme changed — only where the line is drawn.

FIGURE 2

Three years: two programmes improving, one not

Exclusion error, March 2023 to March 2025, on the same definitions. PBI and PIP reach more of their target group each year. PKH does not.

Exclusion error — eligible households not reached

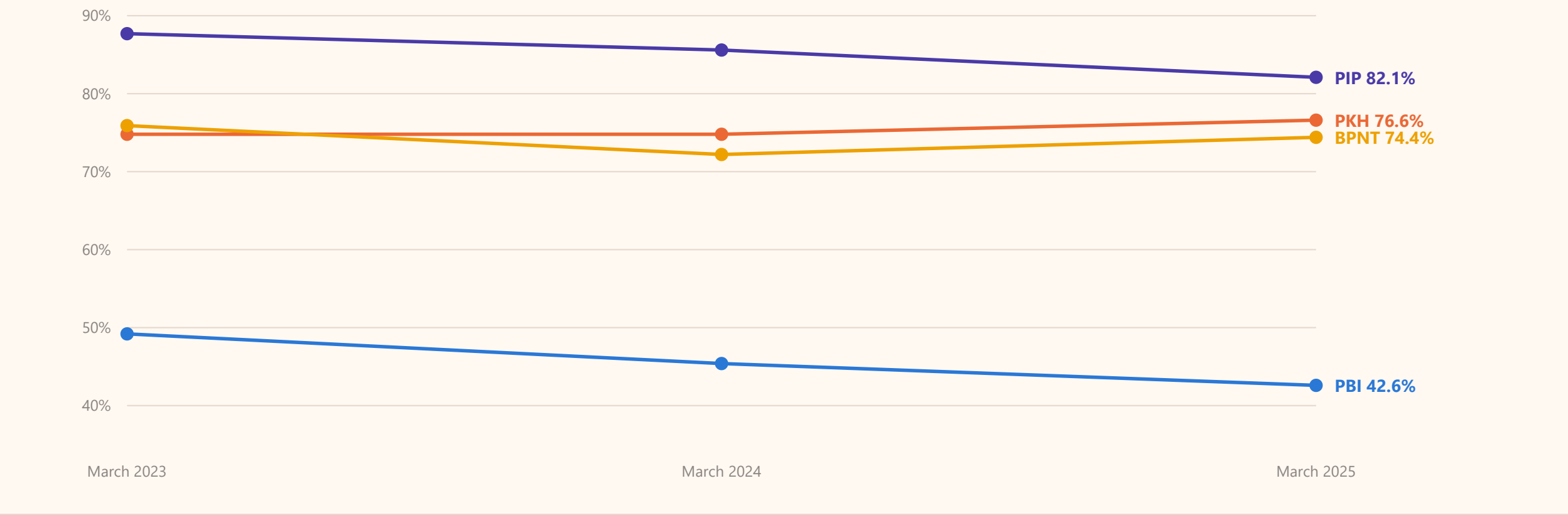


FIGURE 3

The ranking works. The coverage does not.

Share of households or people receiving each programme, by welfare decile. Coverage falls steadily from the poorest decile to the richest in every programme, so the ranking is pointing the right way.

Share receiving the programme

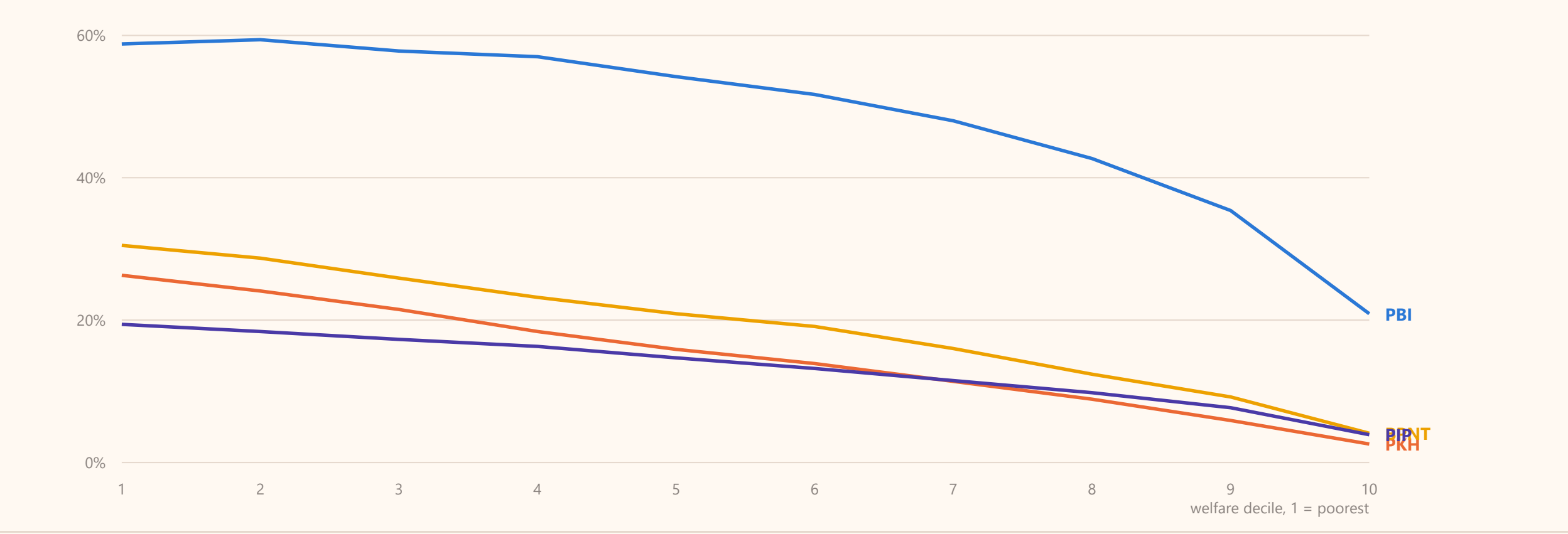
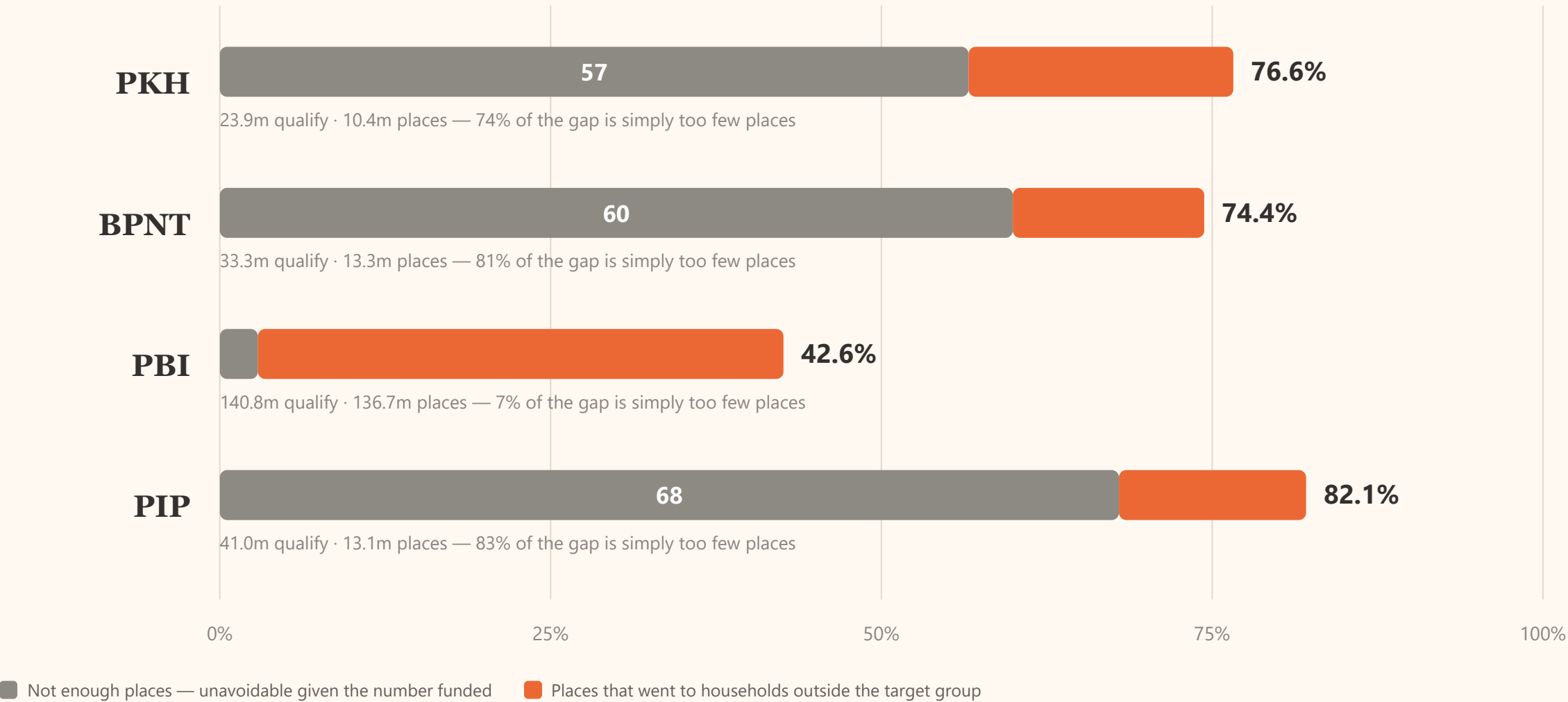


FIGURE 4

How much of the gap is simply too few places

If a programme funds fewer places than there are qualifying households, some of them must be missed however well the targeting works. That floor can be calculated exactly, and it is different for every programme.



For PKH, BPNT and PIP most of the gap is the number of places, and no data system can change that. **PBI is the exception** — it funds almost as many places as there are people who qualify, so nearly all of its 42.6% is places going to the wrong people. That is the part better data can fix.

THE DATA

These numbers are public and can be checked

The Dewan Ekonomi Nasional targeting monitor computes exclusion and inclusion error directly from Susenas microdata, for three years and for every province and district. Anyone can change the target definition and see the numbers move.



Dewan Ekonomi Nasional
Indonesia Social Protection
Targeting Monitor

Target Settings

PKH (Cash Transfer)
Bottom decile
4

☒ Require household conditionality

BPNT (Food Voucher)
Bottom decile
5

PBI (Health Insurance)



Indonesia Social Protection Targeting Monitor BETA
Dewan Ekonomi Nasional

☒ March 2025

☐ March 2024

☐ March 2023

 **National**

EE PKH 76.6%	EE BPNT 74.4%	EE PBI 42.6%	EE PIP 82.1%
IE PKH 46.0%	IE BPNT 36.1%	IE PBI 40.9%	IE PIP 44.2%

EE = Exclusion Error or under-coverage (eligible but not receiving). IE = Inclusion Error or leakage (receiving but not eligible).

Target — PKH: Decile 1–4 + conditionality · BPNT: Decile 1–5 · PBI: Decile 1–5 · PIP: Decile 1–4 + school-age · [Change target settings in the sidebar.](#)

 Incidences

 Province

 District

 Map

 Methodology & Data

Share of population receiving each program by welfare decile — national. Target settings in the sidebar do not apply here.

II Is this unusual?

Other countries publish the same measures. What their experience shows is where the error actually comes from.

FIGURE 5

The same range everywhere

Exclusion and inclusion error for social assistance programmes with published figures. Exclusion error runs from about 50% to 90%, inclusion error from about 24% to 70%. Indonesia sits inside both ranges.

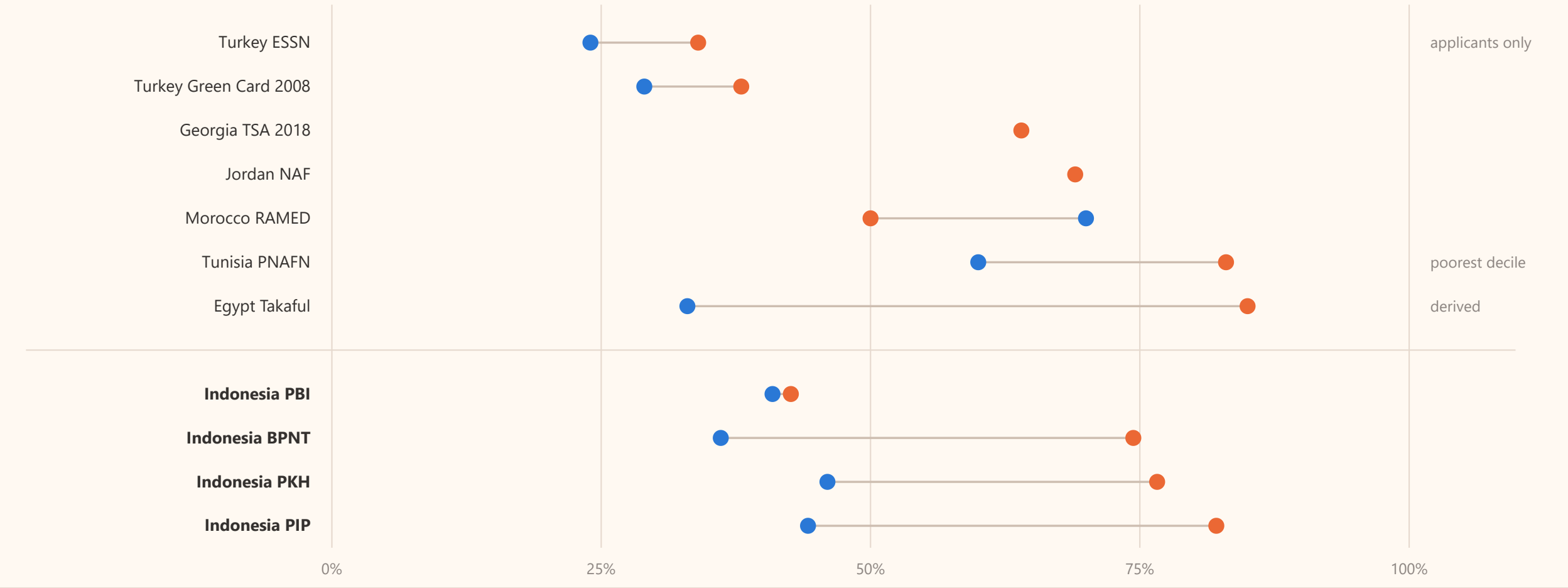
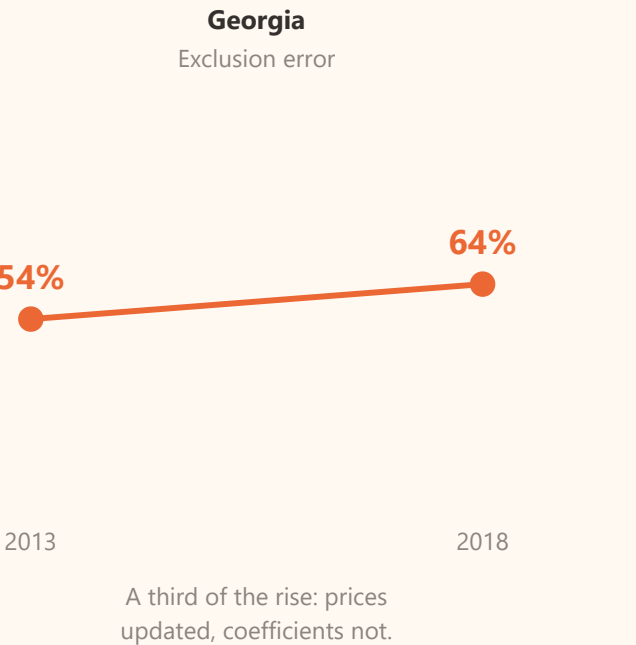
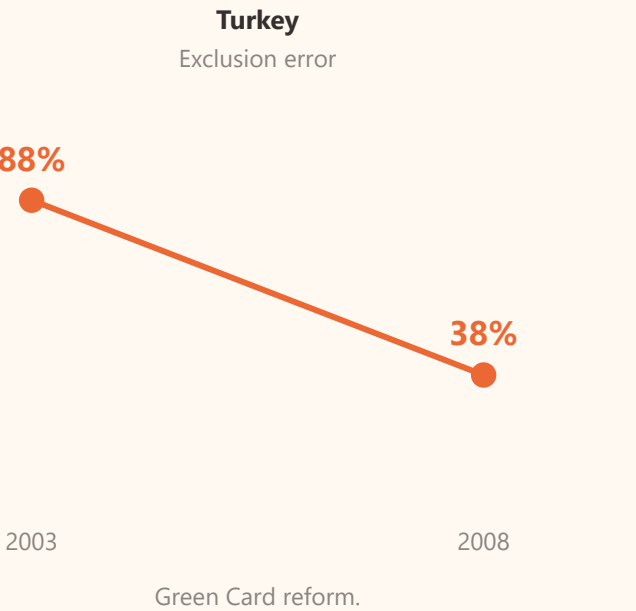
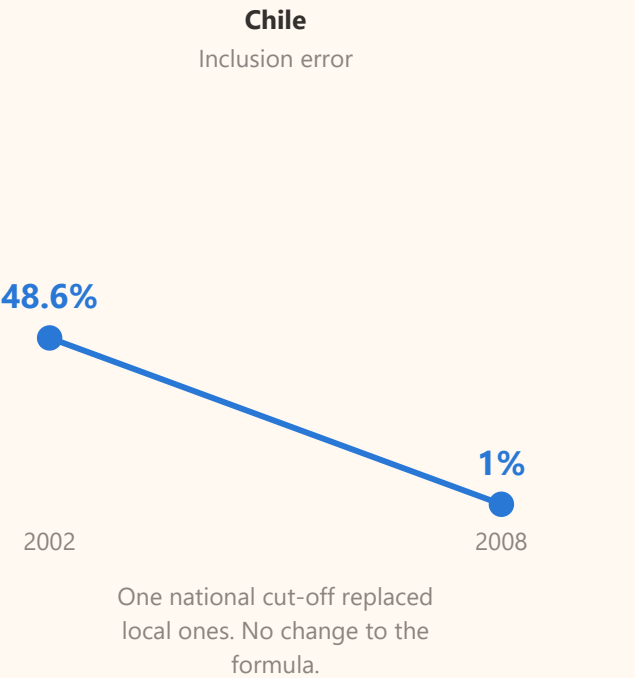


FIGURE 6

What actually changed the numbers elsewhere

In each case where measured error fell sharply, what changed was a rule, a threshold or a process. In none of them was it a better statistical model.



Egypt is the clearest case. Only half of the poorest fifth of households ever applied for Takaful. Researchers calculated that the formula on its own would have admitted 46% non-poor recipients; the figure actually observed was 33%. The programme was more accurate than its formula, because the people who did not need it largely did not apply. The exclusion problem was in **who came forward**, not in the scoring.

III Why is it imperfect?

Two reasons. One is a limit on what any formula can know. The other is everything that happens after the formula.

FIGURE 7

Three ways to decide who is poor

There are three recognised methods. They differ in one thing: how much of a household's income the government can see in its own records.

1

Means test — MT

Measure income directly. Add up income from every source, and test assets, using records the state already holds — tax, payroll, pensions, social security. Nothing is estimated.

Needs: most income visible in records, near-universal formal employment.

Used in: most OECD countries. Chile.

VERIFICATION

2

Proxy means test — PMT

Estimate income from proxies. Collect observable characteristics — housing, assets, household composition, education — and convert them into a score using weights estimated from a survey.

Needs: a household survey and an enumeration round.

Used in: Indonesia, Philippines, Pakistan, Albania, Malawi, Burkina Faso.

ESTIMATION

3

Hybrid means test — HMT

Both at once. Take the income that can be verified from records, estimate only the part that cannot — typically informal earnings — and add them together into one score.

Needs: linked administrative data and a unique ID, plus formal income large enough to be worth verifying.

Used in: Turkey. Common in Eastern Europe.

BOTH

The World Bank does not rank these as better or worse. It presents them as a menu, and says which one fits which situation: a hybrid test “**is recommended when formal income represents a large share of a household’s income**”, and it “**depends on the availability and quality of administrative data**”. That is the criterion that decides what a country can use.

FIGURE 8

How the decile is actually assigned

The survey records what can be observed. A model estimates the relationship between those observations and spending. That model is then applied to 93 million families, and the predicted values are ranked into deciles.

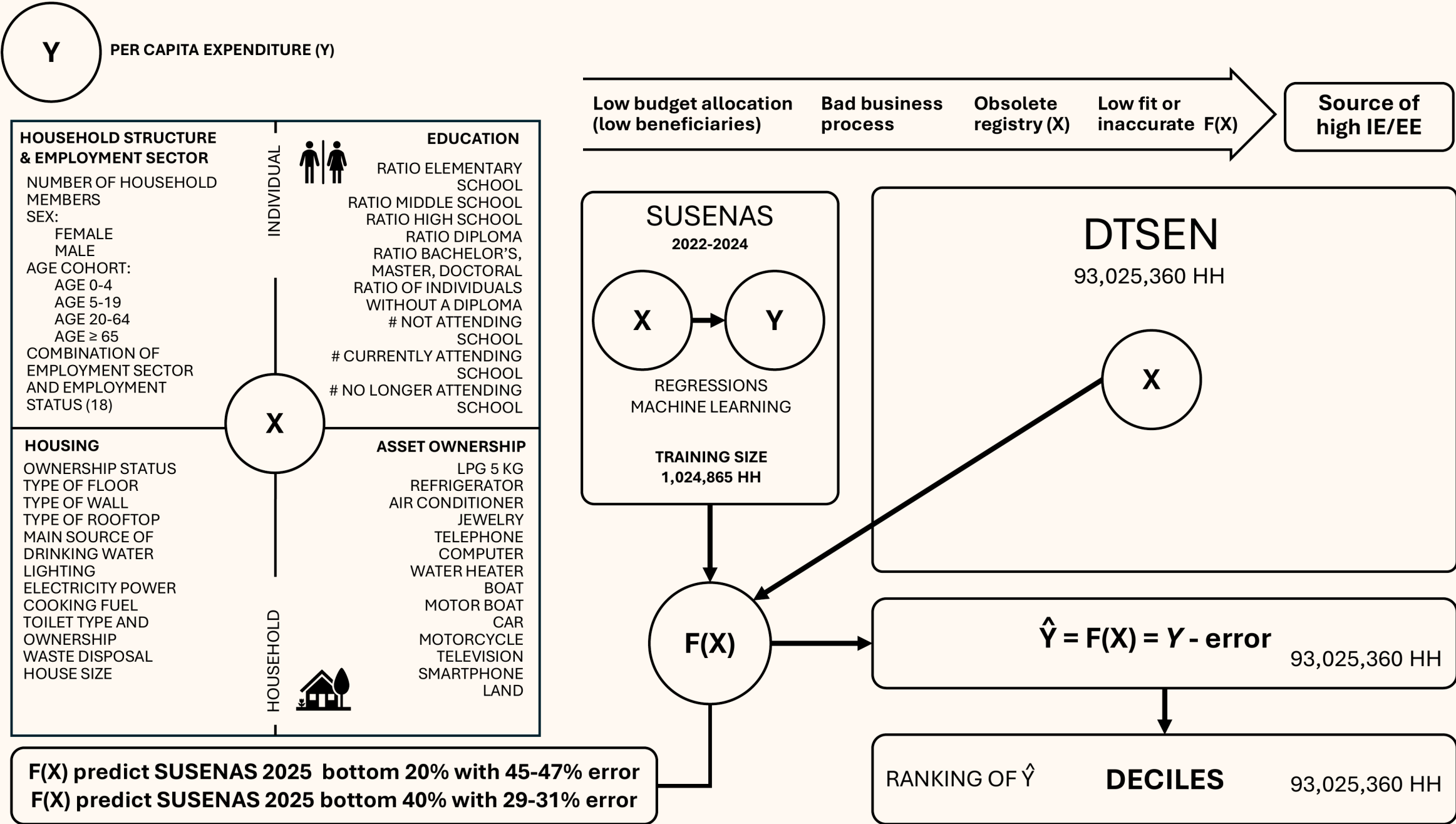


FIGURE 9

What the formula cannot see

The score is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$. $\mathbf{X}$ is everything on the previous slide — housing, assets, schooling, household composition. $\mathbf{e}$ is everything else that decides whether a family is actually poor, and none of it is recorded anywhere.

Poorer than it looks — missed by the formula

Debt. A motorcycle and a refrigerator bought on credit count as assets. The instalments that consume the household's income do not.

Illness. Medical costs for a chronically sick member drain a family for years. The house and the assets look unchanged.

Lost work or a failed harvest. Income falls this season; the roof, the walls and the land are the same as last year.

Care at home. A disabled or elderly member needs a full-time carer, so one adult cannot work.

Supporting relatives elsewhere. Money sent to parents or children in another district leaves the household but never appears in the record.

Richer than it looks — wrongly included

Remittances. A son working in Jakarta or a daughter working abroad sends money home every month. The house still has a dirt floor.

A windfall. An inheritance, a land sale, a lottery win, compensation. Consumption rises; the recorded characteristics do not.

Savings. Cash and gold held at home are worth more than everything on the asset list, and are invisible.

Informal trade. A profitable warung or market stall is income without a payslip and without an asset that gets counted.

Support from adult children living separately but paying the bills.

Two consequences. First, **e runs in both directions**, which is why the same formula produces exclusion error and inclusion error at the same time. Second, **e is large**: the model predicts the bottom 20% with 45–47% error, and published work on the PKH formula puts its R^2 at 0.52 — about half of what determines household spending is outside $\mathbf{X}$. That is also why Indonesia cannot simply copy Chile, which ranks households on recorded income: here, most income is informal and appears in no record.

FIGURE 10

Five sources of error. Only two are the formula.

Errors arise at two separate stages, and each stage produces both kinds of error. In the outcome data they look identical.



Choosing who to help — pensasaran



Delivering the help — penyaluran

1
Out-of-date register

The household's circumstances changed after it was last recorded.

TARGETING

2
Inaccurate formula

The proxies do not capture the household's real situation.

TARGETING

3
Budget limits

The list is cut to the funded quota, whatever the criteria say.

DELIVERY

4
Business process

Records enter unchecked, appeals go unanswered, payments are delayed.

DELIVERY

5
Elite capture

Local influence decides who appears on the list.

DELIVERY

IV What can digitalisation change?

Registration, verification and payment can be improved now. What the formula can know cannot.

FIGURE 11

Self-registration on its own does not work

No country has succeeded with self-service registration alone. Every system where opening registration improved targeting added four things alongside it.

1

Staff to help people apply

19 of 20 on-demand social registries receive applications at staffed local offices. In Surabaya, **84%** of registrations came through an agent and only 16% were self-registered.

INDONESIA: YES

2

Going out to find people

Brazil ran open registration for five years and reached 22 million families, and still found **over one million** more extremely poor families in a single year once it went looking for them.

INDONESIA: PILOTS ONLY

3

Appeals that get answered

An appeal channel corrects the households the formula missed — but only if the appeals are actually decided.

INDONESIA: PARTLY

4

Checking records before accepting them

Administrative data checked at the moment of application, before the record enters the register.

INDONESIA: NO

Opening registration does help. A randomised trial in 400 Indonesian villages found that when households had to come forward and apply, the people who received benefits were **18 to 19% poorer** than under automatic enrolment, and the very poorest were twice as likely to be reached. But only about **60%** of the very poorest applied — and making the process harder screened out poor and rich equally, so any extra difficulty is a cost borne by the poor for no gain.

FIGURE 12

Indonesia's own audit found both ends open

The state audit board examined the register that preceded DTSEN. Records were being accepted without checking, and appeals were being closed without a decision.

224,635

new applications recorded automatically **without verification** — 15,970 of them without the required photographs. BPK: this creates inclusion error.

7,267

appeals closed automatically without being decided — produces exclusion error. Of these, **735 in Jawa Tengah**, 2,085 in Jawa Barat, 758 in Sumatera Utara.

Turkey's answer is simple

In Turkey's integrated system, when an application contains missing or inconsistent information, an alert appears and **the application is frozen until the information is validated**. The applicant cannot proceed and the record cannot enter the register.

Administrative records behind it are refreshed at least every 45 days, and supply about 40% of the information used to score a household. The remaining 60% still comes from a home visit each year.

This is one transferable design feature, not a whole system: **check the record before you accept it, not after.**

CONCLUSION

What digitalisation can and cannot fix

Of the five sources of error, the two that digitalisation closes fastest are not the statistical formula.

SOURCE OF ERROR	CAN DIGITALISATION FIX IT?	
Out-of-date register	Yes, directly	Continuous registration and administrative updates keep the record current.
Business process	Yes, and fastest	Check records at the point of application; decide appeals; pay without delay.
Elite capture	Partly	An audit trail and published lists make local discretion visible.
Inaccurate formula	Only up to a limit	Bounded by how much income can be verified. It improves as formal employment grows, not as the model improves.
Budget limits	No	Not a data problem. Better data changes who receives, not how many.

The digital public infrastructure described earlier today is necessary. This is a map of what it will and will not fix.

Appendix

Reference table, for questions.

Every figure with the population it is measured against

Exclusion and inclusion error can only be compared when you know what each was measured against. These were not measured the same way. The sixth column says how each one differs.

COUNTRY	PROGRAMME	YEAR	EXCLUSION	INCLUSION	MEASURED AGAINST	SOURCE
Turkey	ESSN (refugee cash transfer)	2019	34%	24%	Applicants only — 1.6m of about 3.2m refugees, surveyed before any payment was made. Not comparable with the rest of the table.	Cuevas, Inan, Twose & Çelik (2019), World Bank and WFP
Turkey	Green Card	2003	88%	45%	Poorest decile.	World Bank (2013), Table 3
Turkey	Green Card	2008	38%	29%	Poorest decile. Same programme, five years later.	World Bank (2013), Table 3
Georgia	TSA	2013	54%	—	Bottom quintile of consumption, measured after transfers are removed.	World Bank (2020)
Georgia	TSA	2018	64%	—	As above. About a third of the increase came from prices being updated while the coefficients were not.	World Bank (2020)
Egypt	Takaful	2017	~85%	33%	Exclusion is derived from reported coverage, not stated by IFPRI. Inclusion is measured among beneficiaries, and the population is households with children.	Breisinger et al. (2018), IFPRI
Tunisia	PNAFN	2015	83%	60%	Exclusion against the poorest decile ; inclusion against the poorest quintile . The two are not on the same base.	World Bank (2022)
Tunisia	PNAFN and AMG2	2015	52%	57%	Both against the poorest quintile.	World Bank (2022)
Morocco	RAMED	2019–20	~50%	70%	Exclusion covers RAMED and Tayssir together, against the poorest quintile. Inclusion is the share of beneficiaries outside that quintile.	World Bank (2021)
Jordan	NAF	2010	69%	—	Bottom decile. A second World Bank source reports 83.5% against the poorest quintile — a different denominator, not a contradiction.	World Bank (2012); Silva, Levin & Morgandi (2012)
Chile	Solidario	2002	—	38.6%	Entrants outside the poorest ventile.	Larrañaga & Contreras (2010)
Chile	Solidario	2008	—	1.0%	As above, after local cut-offs were replaced by a single national cut-off. The formula itself did not change.	Larrañaga & Contreras (2010)
Indonesia	PKH	2025	76.6%	46.0%	Deciles 1–4 with programme conditions, households.	Susenas March 2025, own calculation
Indonesia	BPNT	2025	74.4%	36.1%	Deciles 1–5, households.	Susenas March 2025, own calculation
Indonesia	PBI	2025	42.6%	40.9%	Deciles 1–5, individuals.	Susenas March 2025, own calculation
Indonesia	PIP	2025	82.1%	44.2%	Deciles 1–4, school age 6–27, individuals.	Susenas March 2025, own calculation

Sources. Susenas March 2023, 2024 and 2025, own calculation following the definitions of the DEN Indonesia Social Protection Targeting Monitor. Cross-country figures from World Bank and IFPRI publications, each with its own definition of the target population — see the note under each figure. Framework and illustrations from *Salah Sasaran* (Mei 2026). Audit findings from BPK RI, IHPS II 2024.